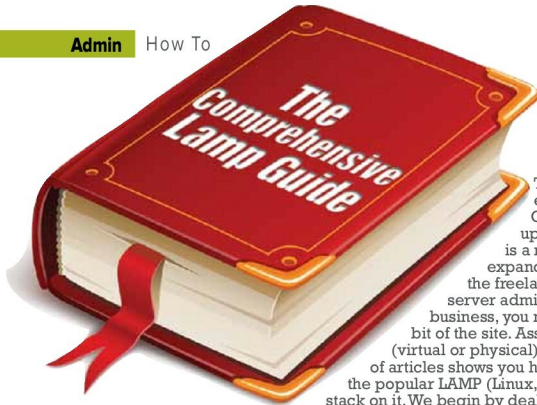




These days, almost everyone needs a website. Online business is picking up, and an Internet presence is a must for firms wanting to expand their business. If you're the freelance Web designer or server administrator for a small-scale business, you need to manage every bit of the site. Assuming you have a server (virtual or physical) running Linux, this series of articles shows you how to install and configure the popular LAMP (Linux, Apache, MySQL and PHP) stack on it. We begin by dealing with Apache.

So what does this article have that the billion others on Google don't? Optimisations.

Most tutorials do not include optimisations, especially compiler optimisations, which increase program efficiency greatly. Note that you won't see the effect of these optimisations unless your server handles over 100 requests per second.

Caveat: Compiler optimisations are not guaranteed to work, but they do in most cases. I have been using them in my set-ups for more than three years without any problems.

Again, the compiler optimisations I use are considered very bad/buggy according to articles on the Internet—but who cares? We want results!

The optimisations used here are highly dependent on the CPU, RAM and CPU cache of the system for which the application is being compiled. Gcc -O3 increases the binary size. This requires a large CPU cache (at least larger than what is required by a binary compiled with -O2). If the cache is small, the application is moved between the CPU cache and DRAM, resulting in application slow down.

We won't be using any of the default/official packages for Apache, MySQL or PHP. Those are optimised for general processors, and do not perform well when compiled specifically for your processor.

In this article, we will not cover the installation of Linux itself, since there are many flavours that are used on servers; your server may have one pre-installed. If not, I assume that you know how to deal with installing Linux and any extra packages, when required.

Installation of Apache (httpd)

As mentioned earlier, there are two things that can be tuned in *httpd*: compiler optimisation and atomic operations. As far as I know, only *httpd* in the LAMP package has this atomic

operations feature, which is used for light-weight thread synchronisation (which must be supported by the CPU, and is the case for most newer processors).

The download

```
$ wget http://apache.inetbridge.net/http/httpd-2.2.17.tar.bz2
```

```
$ wget -O - http://www.apache.org/dist/http/httpd-2.2.17.tar.bz2.sha1 | sha1sum -c && { tar -xzf httpd-X.Y.Z.tar.bz2; cd httpd-X.Y.Z; }
```

The *apache.inetbridge.net* host-name can be replaced by any other official mirror that distributes the Apache source code. These commands download the tarball for Apache. Next, check for the integrity of the downloaded file with the source checksum from Apache's official mirror, ensuring the code doesn't contain modifications not known to Apache developers.

Compilation settings in the environment

The compilation method for almost every C program distributed for the Linux family is to use GNU *autotools*. Yes, there are some variations—in our case, the latest release of MySQL uses Cmake—but we'll deal with that in the article on installing MySQL. Usually, Cmake is used for applications written in C++, but is not constrained to these.

The *configure* script generated when developers use GNU *autotools* recognises certain environment variables that need to be passed to relevant tools: the compiler/linker, etc.

The *CFLAGS* environment variable can contain extra options that need to be passed to the C compiler. We set this as follows: